

Marcus Sostak

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EDUCATION

University of South Carolina

Columbia, SC

Bachelor of Science in Computer Science, Minor in Mathematics | **GPA: 3.92**

May 2028

- **Relevant Coursework:** Software Engineering, Data Structures & Algorithms, Advanced Programming (C++), Operating Systems, Computer Architecture
- **Organizations:** Gamecock AI/ML Club, Kappa Theta Pi Professional Technical Fraternity

EXPERIENCE

Undergraduate Research Assistant

May 2025 – Present

University of South Carolina

Columbia, SC

- Developed and maintained an educational web platform used by local elementary schools for interactive word-based learning using RedwoodJS, React, GraphQL APIs and PostgreSQL.
- Refactored legacy frontend and backend components, removing 1,000+ lines of redundant code and improving maintainability, stability, and deployment reliability.
- Implemented automated preview environments integrated with CI, reducing code review turnaround time by $\sim 50\%$ and decreasing post-deployment bugs.
- Collaborated with a faculty mentor and a team of 4 developers on a \$250K+ grant-funded research project, using Trello to plan Agile sprints, prioritize features, and ship user-driven improvements.
- Authored technical and onboarding documentation and proposed a roadmap for integrating AI-driven personalization and content generation based on user interaction data.

PROJECTS

Utility Pole Inspection AI — 1st Place GridStorm Hackathon | *Python, YOLO, PaddleOCR, Roboflow*

- Fine-tuned two YOLOv26 models on a custom dataset of 1,500+ field images (7-class detection, 2-class segmentation), achieving 0.84 mAP@0.5 after data augmentation.
- Applied checkpoint weight averaging (“model soup”) across 4 training checkpoints, improving validation mAP by +3.1% and reducing per-class variance without increasing inference latency.
- Built a multi-variant OCR pipeline (original + CLAHE-enhanced) using PaddleOCR, improving plate-level read accuracy on noisy field images via per-character confidence aggregation and regex validation.
- Designed a geometry-based encroachment scoring method using vegetation-mask overlap ratios.

Climbing Gym Hold Detection | *PyTorch, Matplotlib, NumPy, Computer Vision*

- Engineered a CNN using PyTorch to classify 6 indoor climbing hold types, leveraging CUDA-accelerated training to iterate on architecture and preprocessing for near real-time inference.
- Architected an end-to-end data processing pipeline to download data, crop holds using YOLO annotations, and pad images to 128×128 , producing 60,000 labeled samples.
- Visualized model robustness across viewing environments using Matplotlib epoch graphs and confusion matrices.
- Optimized video processing workflows using a YOLOv11 detector, achieving 92% accuracy on hold detection and supporting classification at 30 FPS, exploring route analysis and movement tracking.

Dialect Classifier App | *React, FastAPI, Supabase, TailwindCSS, JavaScript, Docker*

- Deployed a full-stack web application for automatic dialect classification using React + Vercel on the frontend and a Dockerized FastAPI backend on Render for real-time data handling.
- Coded backend services for audio capture, metadata persistence, and RESTful APIs using Supabase, enabling collection of 1000+ labeled ML training samples while implementing clear user consent disclosures.
- Prototyped a modular, scalable backend architecture to support preprocessing, feature extraction, and future training of acoustic or embedding-based classifiers, enabling experimentation and rapid iteration.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript

Frameworks & Libraries: PyTorch, YOLO, OCR, NumPy, Scikit-learn, Matplotlib, Seaborn, React, Flask

Developer Tools: Git, GitHub, Linux, Jupyter, Docker, AWS (S3), Render, Vercel

Databases & APIs: PostgreSQL, MySQL, Supabase, GraphQL, REST